Vishal Bhat

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Education

R. V. College of Engineering (RVCE) Bangalore, India
Bachelor of Engineering in Computer Science and Engineering; CGPA: 9.78 *Aug. 2023 – Present*

Experience

Software Engineering Intern May 2026 – Jul. 2026
LinkedIn *Bangalore, India*

- Worked on the Experimentation Platform Infrastructure team, designing and delivering a new end-to-end feature for a high-scale internal platform used across LinkedIn engineering workflows.
- Implemented production-grade gRPC services in a Java Spring Boot backend, enabling efficient service-to-service communication within distributed infrastructure components.
- Developed data access and integration layers connecting Apache Pinot, HDFS, and SQL-based storage systems to support low-latency experimentation and analytics workflows.
- Built and integrated a Copilot MCP plugin and harness framework, enabling AI-assisted developer workflows and extensible tooling within internal engineering platforms.
- Integrated machine learning model prediction pipelines into backend services, enabling real-time inference and feature-driven decision workflows across internal infrastructure systems.

PRISM Research Intern – Samsung PRISM Jun. 2025 – Nov. 2025
Samsung R&D Institute India (SRIB) *Bangalore, India*

- Contributed to SmartThings AI Search, fine-tuning vector embedding pipelines and quantized Mini LLMs (TinyLlama, Phi-2) to accurately extract intent from complex natural language queries across 50+ device actions.
- Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline leveraging FAISS and ChromaDB to dynamically index SmartThings device capabilities, enabling low-latency, context-aware command generation.
- Engineered a dynamic prompt orchestration module and built custom semantic re-ranking logic with SRIB mentors, improving device action suggestion accuracy by 18% over baseline keyword models.
- Delivered production-ready REST API integrations connecting local LLM inference outputs directly to SmartThings execution endpoints, achieving near real-time device response and command verification.

Software Developer – Journal Publishing Portal Aug. 2024 – Nov. 2024
R. V. College of Engineering (for ASM India) *Bangalore, India*

- Built and deployed a full-stack journal portal using React, Node.js, and Firebase, supporting 30+ authors and 75+ research articles across the submission and peer-review lifecycle.
- Implemented Firebase Authentication, Firestore, and Cloud Storage with role-based access for 3+ user roles, managing article submissions, metadata, reviews, and editorial decisions.
- Collaborated with 10+ faculty members, editors, and stakeholders to gather requirements and translate publishing workflows into 15+ functional features and interfaces.
- Streamlined article management and peer-review coordination through a centralized platform, reducing manual handling across the journal's submission-to-publication workflow.

Projects

LLM IR Repair Architecture | *Python, LLVM, C++, LLMs*

- Built a research framework evaluating LLM-generated LLVM IR across an 18-case benchmark covering arithmetic, control flow, loops, function calls, and SSA phi-node lowering.
- Engineered a staged validation and bounded repair pipeline using `llvm-as`, `opt`, and `lli` to detect syntax, structural, and semantic failures and feed compiler diagnostics back to LLMs for iterative repair.

NeuralShield: Malware Detection Tool | *Python, Scikit-learn, Flask, React*

- Built a machine-learning malware detector using Random Forest on PE header features, achieving 94.37% accuracy and 0.94 F1-score on the evaluation dataset.
- Engineered a feature-extraction and selection pipeline with `pefile` and Scikit-learn to reduce feature noise while maintaining a 10.5% train-test performance gap.

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript, Java, HTML, CSS, SQL
AI/ML & RAG: LLMs (TinyLlama, Phi-2), RAG, FAISS, ChromaDB, Vector Embeddings, Scikit-learn, NumPy, Pandas
Web Technologies: MERN Stack (MongoDB, Express.js, React, Node.js), Spring Boot, REST APIs, Flask, Tailwind CSS
Databases & Messaging: MongoDB, PostgreSQL, Redis, Firestore, Vector Databases (FAISS, ChromaDB), Apache Kafka
Developer Tools: Git, GitHub, VS Code, Postman, Netlify, Heroku, Vercel, Render, SmartThings API